Predicting 30-Day Hospital Readmission for Diabetic Patients: A Comparative Analysis of Machine Learning Classifiers

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Abstract. Unplanned 30-day hospital readmissions among diabetic patients cost the U.S. healthcare system billions annually, yet predicting which patients will return remains difficult. We trained five classifiers—decision tree, logistic regression, linear SVM, XGBoost, and a feedforward neural network—on the Diabetes 130-US Hospitals dataset (101,766 encounters from 130 hospitals) after patient-level deduplication and leakage-aware preprocessing that reduced the cohort to roughly 70,000 rows and 98 features. All models used consistent class-weight adjustments for the 9% positive rate. XGBoost achieved the highest test AUC-ROC at 0.6505, but the gap separating the best from the worst model was only 0.035—confirming that the readmission signal extractable from administrative data alone is weak. SHAP analysis identified discharge disposition and prior inpatient visits as the dominant predictors across all methods. A fairness audit by race revealed recall disparities that persisted across model families, pointing to data-level rather than model-level inequity. We argue that threshold selection, calibration, and demographic monitoring matter at least as much as which algorithm a hospital picks.

Keywords: Hospital readmission · Diabetes · XGBoost · SHAP · Fairness · Classification

1 Introduction

Thirty-day hospital readmissions cost the United States roughly $26 billion per year, and diabetic patients are readmitted at disproportionately high rates [6]. Since 2012, the Centers for Medicare and Medicaid Services (CMS) have penalized hospitals with excess readmission rates under the Hospital Readmissions Reduction Program (HRRP), turning what was already a clinical problem into a financial one [7]. Any tool that can flag high-risk patients before discharge—even imperfectly—creates an opening for targeted interventions: medication reconciliation, follow-up calls within 48 hours, or transitional care coordination.

Machine learning is attractive here because a clinician cannot mentally integrate dozens of features at once—lab counts, medication changes, diagnosis

categories, discharge disposition codes—while a classifier can. The hope is that nonlinear interactions between these variables carry predictive signal that simpler risk scores miss. Whether that hope is justified for *this particular dataset* is exactly what our experiments test.

We trained five classifiers on the Strack et al. (2014) Diabetes 130-US Hospitals dataset [6]: a pruned decision tree, L2-regularized logistic regression, a calibrated linear SVM, XGBoost with randomized hyperparameter search, and two feedforward neural network architectures (keeping the better one). Every model received the same stratified 80/20 split, the same class-weight correction for the 9% positive prevalence, and the same 98-feature matrix produced by a leakage-aware preprocessing pipeline. We evaluated primarily on AUC-ROC, added SHAP-based explanations for XGBoost, performed threshold optimization, and ran a fairness audit sliced by race.

Our contributions are fourfold. First, we enforce patient-level deduplication and train-only scaling to prevent data leakage—a step that some prior work on this dataset omits. Second, we provide a controlled head-to-head comparison where the only variable is the learning algorithm. Third, we demonstrate that threshold selection can matter more than model selection (the SVM produced exactly one positive prediction at a 0.5 cutoff until we adjusted). Fourth, we report demographic fairness slices alongside performance metrics, because a model that is accurate on average but systematically misses readmissions in underrepresented groups is not ready for deployment.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset, preprocessing, and feature engineering. Section 4 details each classifier. Section 5 presents results, and Section 6 offers interpretation, limitations, and recommendations. Section 7 concludes.

2 Related Work

Strack et al. [6] released the Diabetes 130-US Hospitals dataset and studied the relationship between HbA1c measurement and readmission, concluding that HbA1c testing was associated with lower readmission rates. Their analysis was primarily statistical rather than predictive; they did not train or compare classifiers in the modern ML sense. The dataset has since become a benchmark, but many subsequent papers treat it as a straightforward classification problem without acknowledging how weak the signal really is.

Emi-Johnson et al. [3] recently applied XGBoost to this same cohort and reported an AUC of approximately 0.667. Our pipeline yields 0.6505—a difference we attribute to choices in deduplication, feature filtering, and random seeds rather than to a methodological flaw. The gap actually reinforces an important point: on this dataset, small implementation decisions move AUC by a couple of hundredths, which means reported differences between algorithms of similar magnitude should not be over-interpreted.

Rajkomar et al. [5] demonstrated that deep learning on electronic health records can outperform traditional models when given access to rich temporal

Table 1. Feature taxonomy of the raw dataset, grouped by clinical domain.

Domain	Features (examples)	Count
Demographics	age, gender, race	3
Encounter admin	admission_type, source, discharge_disposition	3
Utilization	time_in_hospital, num_lab_procedures, num_medications	6
Prior history	number_inpatient, number_outpatient, number_emergency	3
Diagnoses	diag_1, diag_2, diag_3 (ICD-9 codes)	3
Lab results	max_glu_serum, A1Cresult	2
Medications	metformin, insulin, glipizide, ... (23 drug columns)	23
Other/admin	patient_nbr, encounter_id, weight, payer_code, ...	7

data (sequences of clinical events over time). Our setting is different: we have a single flat row per encounter with no temporal ordering, which removes the advantage that recurrent or attention-based architectures would otherwise enjoy. Bayati et al. [1] explored penalized regression for readmission prediction and found that L1-regularized models performed comparably to more complex alternatives when feature engineering was done carefully—a finding our logistic regression results echo.

On the fairness side, Obermeyer et al. [4] showed that a widely deployed commercial algorithm for identifying high-risk patients systematically underestimated the health needs of Black patients. While our fairness audit is descriptive rather than causal, it is motivated by the same concern: a model that works well "on average" can hide group-level disparities that matter clinically.

We position our work as an honest replication-plus-extension exercise. We do not claim a new state-of-the-art. Instead, we add the dimensions of fairness analysis and threshold-awareness to a benchmark that is frequently used but rarely interrogated critically.

3 Dataset and Methodology

3.1 Dataset Description

The Diabetes 130-US Hospitals dataset (UCI Machine Learning Repository, ID 296) contains 101,766 inpatient encounters from 130 U.S. hospitals between 1999 and 2008 [6]. Each row represents one hospital encounter for a patient whose stay involved a diabetes diagnosis. The original feature set includes 50 columns spanning demographics (age bracket, gender, race), admission metadata (type, source, discharge disposition), clinical measures (number of lab procedures, number of medications, HbA1c and glucose test results), ICD-9 diagnosis codes for three diagnosis slots, and 23 binary medication-change indicators. Table 1 summarizes the feature groups.

The binary target, `readmitted_binary`, equals 1 if the patient was readmitted within 30 days and 0 otherwise. The positive rate is approximately 9%.

Table 2. Data flow from raw encounters to final feature matrices.

Stage	Row / Feature Count	Key Operation
Raw download	101,766 × 50	—
Patient dedup	71,518	Keep first encounter per patient
Cleaning	∼70,000	Remove deceased/hospice, invalid gender
Encoding	∼70,000 × 121	One-hot ICD-9 groups, race, admin codes
Feature engineering	∼70,000 × 132	Add utilization aggregates, flags
Variance + VIF filter	∼70,000 × 98	Drop NZV columns, high-multicollinearity
Train / test split	55,976 / 13,994	80/20 stratified

3.2 Preprocessing

Our preprocessing pipeline enforces three safeguards against data leakage, applied in sequence before any model sees the data.

Patient deduplication. A single patient may appear multiple times across encounters. We kept only the first encounter per `patient_nbr`, reducing 101,766 rows to 71,518. Without this step, training on one visit and testing on a later visit from the same patient inflates performance estimates.

Cleaning. We removed rows with invalid gender entries, deceased patients, and those discharged to hospice—outcomes where readmission is either undefined or irrelevant. Columns with more than 40% missingness (e.g., weight, payer code) were dropped. ICD-9 codes in the three diagnosis slots were mapped to approximately 10 clinical groups (Circulatory, Diabetes, Respiratory, Musculoskeletal, etc.) and one-hot encoded. Race and admission-related categoricals were also one-hot encoded. Sixteen near-zero-variance medication columns were removed.

Train-only scaling. We split 80/20 stratified by the target (`random_state=42`), yielding 55,976 training rows and 13,994 test rows. A `StandardScaler` was fit exclusively on the training fold and applied to both folds. Unscaled copies were retained for tree-based models.

After these steps, the feature matrix contained 98 columns. Table 2 summarizes the pipeline.

3.3 Feature Engineering

We constructed several domain-motivated features before the variance filter. `total_visits_prior` sums outpatient, inpatient, and emergency visits. `n_med_changes` counts how many of the 23 medication columns changed (up or down). `a1c_tested` and `glu_tested` are binary flags indicating whether HbA1c or glucose were measured at all. `stay_short`, `stay_medium`, and `stay_long` bucket length of stay into clinically interpretable categories. `number_outpatient_log` applies a log transform to the right-skewed outpatient count.

After engineering, a variance threshold removed columns with near-zero variance, and a multicollinearity check pruned features whose variance inflation factor exceeded a conservative cutoff. The final matrix has 98 features.

3.4 Experimental Setup

All models were evaluated on the same held-out 13,994 test rows. Imbalance handling was consistent: `class_weight='balanced'` for scikit-learn models, `scale_pos_weight`=10.15 for XGBoost, and an equivalent `class_weight` dictionary for Keras. Hyperparameter tuning used 5-fold stratified cross-validation on the training set: `GridSearchCV` for the decision tree, logistic regression, and SVM; `RandomizedSearchCV` (18 iterations) for XGBoost; and early stopping on a 15% validation split for neural networks.

Our primary metric is AUC-ROC because it is threshold-free and appropriate for imbalanced problems. We also report accuracy, precision, recall, and F1 at the default 0.5 cutoff to illustrate how misleading those metrics can be when the positive class is rare.

4 Models

4.1 Decision Tree

The decision tree serves as our simplest baseline. It partitions the feature space with axis-aligned Gini-impurity splits, requires no scaling, and produces human-readable rules. We searched over `max_depth` $\in \{3, 5, 7, 10, 15, \text{None}\}$, `min_samples_split` $\in \{2, 50, 120\}$, and `min_samples_leaf` $\in \{1, 20, 60\}$. The best configuration was depth 7 with `min_samples_leaf`=60, a conservative prune that sacrifices some training accuracy for stability.

4.2 Logistic Regression

Logistic regression estimates a linear decision boundary and outputs calibrated probabilities via the logistic function. We tuned the L2 penalty strength C over 13 values from 10^{-3} to 10^{3} using the LBFGS solver. The optimal C was 0.001—heavy regularization—which tells us the model prefers shrinking most coefficients close to zero rather than fitting every feature aggressively. The resulting coefficients are directly interpretable as log-odds per standardized unit, which makes this model the easiest to audit.

4.3 SVM (Linear)

We used `LinearSVC` to learn a maximum-margin hyperplane. A full RBF-kernel SVM on ∼56,000 rows is $O(n^2)$ in memory and impractical on commodity hardware, so we stayed linear. After tuning C via grid search (best: 0.001), we wrapped the selected estimator in `CalibratedClassifierCV` (sigmoid method, 3-fold) to obtain probability estimates for AUC computation. An important lesson emerged: at the default 0.5 threshold, the calibrated model produced exactly *one* positive prediction on 13,994 test rows. The AUC was still 0.6442—the ranking was fine, but the threshold was not. We adjusted it to 0.108 using F1 optimization on training probabilities.

Table 3. Test-set performance of all five classifiers. Precision, recall, and F1 refer to the positive class (readmitted within 30 days) at a 0.5 probability threshold.

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Decision Tree	0.5864	0.1214	0.5793	0.2008	0.6152
Logistic Regression	0.6577	0.1347	0.5195	0.2140	0.6444
SVM (Linear)	0.7524	0.1581	0.4072	0.2278	0.6442
XGBoost	0.6621	0.1403	0.5394	0.2226	0.6505
Neural Net	0.6934	0.1455	0.4964	0.2251	0.6443

4.4 XGBoost

XGBoost [2] fits shallow decision trees sequentially, each correcting the residual errors of the ensemble so far. L1/L2 regularization on leaf weights and row/column subsampling control overfitting. We searched 18 random configurations over `n_estimators` $\in \{100, 300, 500\}$, `max_depth` $\in \{3, 5, 7\}$, `learning_rate` $\in \{0.01, 0.1, 0.3\}$, `subsample` $\in \{0.7, 0.9, 1.0\}$, and `colsample_bytree` $\in \{0.7, 0.9, 1.0\}$. The winner used 500 trees at depth 5 with a learning rate of 0.01 and both subsampling ratios at 0.7—a high-regularization configuration.

4.5 Neural Network

We built two feedforward architectures on the scaled inputs. The basic network has a single 64-unit ReLU hidden layer with 30% dropout. The deeper network stacks 128, 64, and 32 units with batch normalization and dropout at each stage, trained with a slower Adam learning rate (0.0005) and a reduce-on-plateau schedule. Both use sigmoid output, binary cross-entropy loss, and early stopping on validation loss.

The basic network stopped at epoch 15 (out of 50). The deep network ran all 50 epochs before the learning rate was halved. On the test set, the basic architecture achieved 0.6443 AUC versus 0.6383 for the deeper one, so we kept the basic network for the final comparison. We were not surprised: ∼56,000 rows and 98 pre-engineered features is not the regime where deep learning typically outperforms gradient boosting.

5 Results

5.1 Model Comparison

Table 3 presents test-set performance for all five models. XGBoost leads on AUC-ROC at 0.6505, followed by logistic regression (0.6444), the neural network (0.6443), and the linear SVM (0.6442)—these three are essentially tied. The decision tree trails at 0.6152.

The total AUC spread across all five models is 0.035. Fig. 1 shows the overlaid ROC curves: they cluster tightly, separating from the diagonal but never

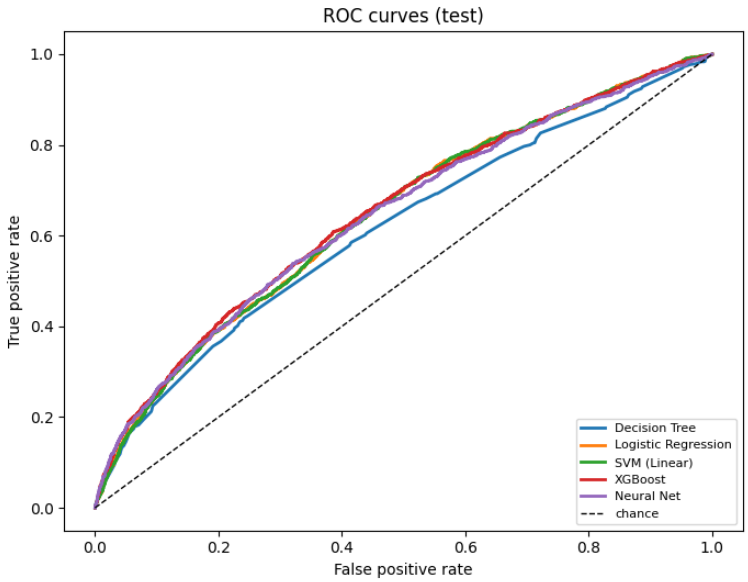

Fig. 1. ROC curves for all five classifiers on the held-out test set. The curves cluster between AUC 0.615 and 0.651, confirming that model family matters less than the inherent difficulty of the prediction task.

approaching the upper-left corner. The readmission signal in this data is, frankly, weak.

Fig. 2 compares AUC-ROC and F1 side by side. The F1 scores are all low (0.20–0.23) because precision at the 0.5 cutoff is punished by the 9% base rate: most positive predictions are false alarms.

5.2 Overfitting Analysis

Table 4 reports train AUC alongside test AUC. Logistic regression and the linear SVM exhibit gaps of only 0.013, while the deep neural network shows a 0.156 gap—clear overfitting. XGBoost sits in between at 0.066, and the basic neural network at 0.079. The decision tree's gap of 0.049 is moderate, reflecting the benefit of pre-pruning.

5.3 Feature Importance

Fig. 3 shows the SHAP beeswarm plot for XGBoost on the test set. The feature `discharge_disposition_id_1` ("discharged to home") dominates, with a mean absolute SHAP value of 0.227—more than double the second-ranked feature. Patients discharged home have a *lower* predicted risk (the high feature values push SHAP negative), which makes clinical sense: being sent home signals clinical stability, whereas transfer to another facility or skilled nursing suggests complexity.

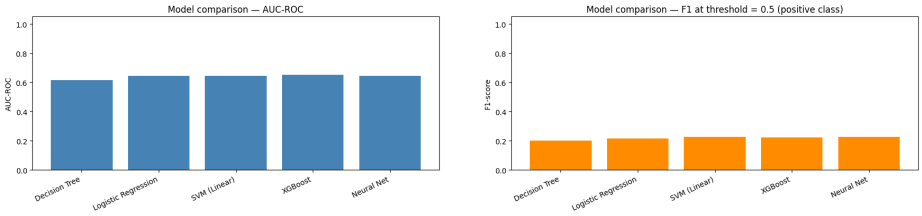

Fig. 2. Left: AUC-ROC comparison. Right: F1-score at threshold=0.5. The F1 bars are uniformly low because precision collapses at the default cutoff with 9% prevalence.

Table 4. Train vs. test AUC-ROC and the gap between them. Larger gaps indicate overfitting.

Model	Train AUC	Test AUC	Gap
Decision Tree	0.6645	0.6152	0.0493
Logistic Regression	0.6575	0.6444	0.0131
SVM (Linear)	0.6575	0.6442	0.0133
XGBoost	0.7168	0.6505	0.0663
Neural Net (basic)	0.7236	0.6443	0.0793
Neural Net (deep)	0.7947	0.6383	0.1564

Table 5 compares the top-10 features across four methods. Discharge disposition and prior inpatient visits appear at the top of every list. This cross-method convergence is reassuring: it suggests these signals are genuine properties of the data rather than artifacts of a particular algorithm's inductive bias.

5.4 Threshold Optimization

We swept probability thresholds from 0.05 to 0.55 on a stratified 15% hold-out carved from the training set, optimizing F1 for the positive class. Fig. 4 shows the curve. The optimum landed at 0.515—barely different from 0.5. On the test set, F1 rose from 0.2226 to 0.2257, a marginal gain. The near-optimal default is a sign that XGBoost's internal probability estimates are already reasonably calibrated for this problem.

Contrast this with the linear SVM, where the default 0.5 cutoff produced a single positive prediction out of 13,994 test rows. After shifting the threshold to 0.108, the SVM recovered 511 true positives. The SVM's AUC was fine (0.6442); the problem was entirely about the operating point. This example makes the general lesson concrete: AUC measures ranking quality, but deployment requires an explicit cutoff, and that cutoff can make or break a model's clinical usefulness.

5.5 Fairness Analysis

Table 6 reports recall and F1 by reconstructed race group for both XGBoost and logistic regression. The "Other" group has the lowest recall under both models

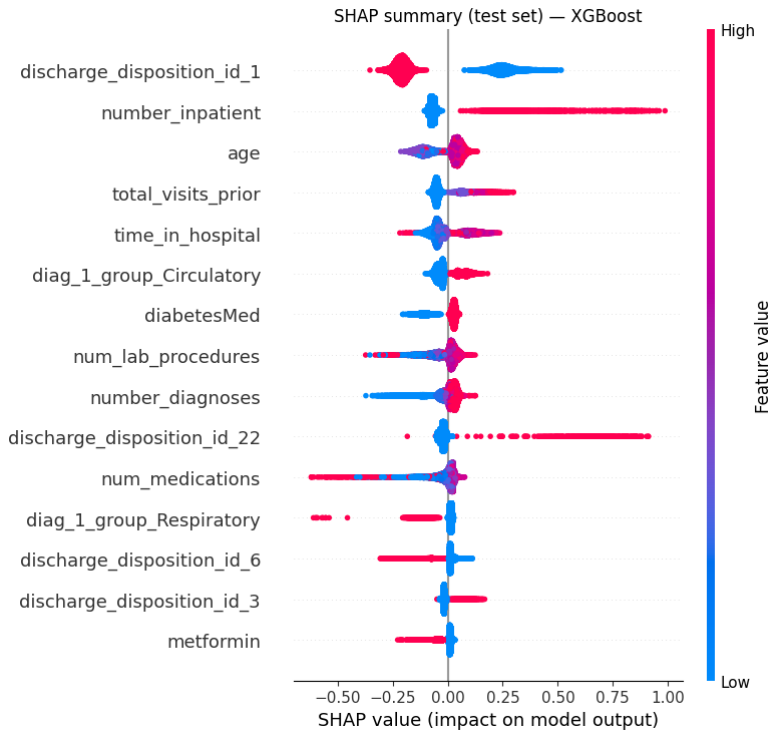

Fig. 3. SHAP beeswarm summary for XGBoost (test set). Color indicates feature value (red=high, blue=low); horizontal position indicates SHAP contribution to the predicted log-odds of readmission.

(0.421), while the Hispanic group has the highest (0.571), though that slice contains only 291 patients and the estimate is noisy.

The key observation is that the gaps between groups are nearly identical across both models (Fig. 5). XGBoost and logistic regression have entirely different inductive biases, yet they produce the same pattern of group-level disparities. This consistency suggests that the disparity lives in the data—in documentation practices, access patterns, and baseline prevalence differences across groups—rather than in what the model learns.

6 Discussion

XGBoost achieved the best test AUC at 0.6505, but we want to be clear about what that means. The margin over logistic regression is 0.006. On five-fold cross-validation, these differences are within the range of noise. Our result aligns with published benchmarks on this dataset: Emi-Johnson et al. [3] report approximately 0.667, and the gap between their number and ours is easily explained by

Table 5. Top-10 features by importance rank across four methods. Boldface marks features appearing in all four lists.

Rank	Decision Tree	Logistic Reg.	XGBoost	SHAP (XGB)
1	**disch_disp_1**	**disch_disp_1**	**disch_disp_1**	**disch_disp_1**
2	**num_inpatient**	**num_inpatient**	disch_disp_22	**num_inpatient**
3	disch_disp_22	disch_disp_22	**num_inpatient**	age
4	num_lab_proc	disch_disp_5	disch_disp_3	total_visits_prior
5	num_medications	diabetesMed	total_visits_prior	time_in_hospital
6	diag1_Circulatory	num_diagnoses	diag1_Musculoskel	diag1_Circulatory
7	age	disch_disp_3	disch_disp_5	diabetesMed
8	diag1_Musculoskel	disch_disp_25	stay_short	num_lab_proc
9	num_diagnoses	age	diag1_Circulatory	num_diagnoses
10	disch_disp_6	diag2_Neoplasms	disch_disp_6	disch_disp_22

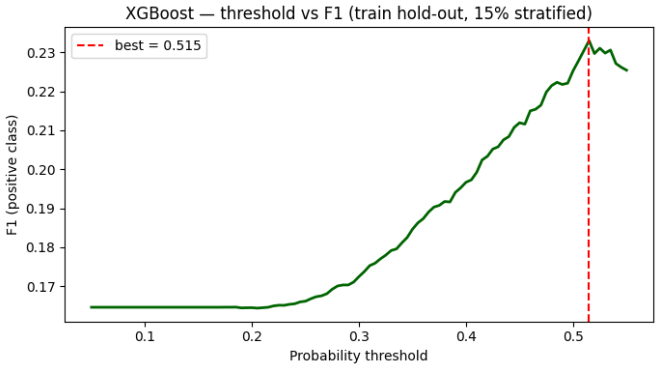

Fig. 4. F1-score (positive class) as a function of probability threshold for XGBoost, evaluated on a held-out 15% stratified slice of the training data.

differences in deduplication, feature selection, and random seeds. The readmission signal extractable from administrative encounter data is weak and noisy. No algorithm we tested—or that we have seen published—pushes AUC much beyond 0.67.

Why does gradient boosting fit this problem at all? Each shallow tree corrects the mistakes of the ensemble so far, gradually accumulating evidence from many partial rules. Regularization (L1/L2 on leaf weights) and stochastic subsampling (70% of rows and 70% of columns per tree) prevent the ensemble from memorizing training noise. The train–test AUC gap for XGBoost was 0.066—nontrivial, but far smaller than the deep neural network's 0.156 gap. Boosting trades a small amount of bias for a substantial reduction in variance relative to a single deep tree or a high-capacity neural network.

The neural network results deserve honest commentary. We gave the deeper architecture batch normalization, dropout, a slow learning rate, reduce-on-plateau

Table 6. Fairness slices: recall and F1 by race for XGBoost and logistic regression (test set, 0.5 threshold). n is the number of test-set patients in each group.

		XGBoost		**Logistic Reg.**	
Race	n	Recall	F1	Recall	F1
African American	2,544	0.513	0.228	0.504	0.218
Caucasian	10,852	0.547	0.221	0.523	0.212
Hispanic	291	0.571	0.294	0.571	0.314
Other	307	0.421	0.182	0.421	0.178

Fig. 5. Recall and F1 by race for XGBoost and logistic regression. The pattern of disparities is consistent across both model families, suggesting a data-level rather than model-level origin.

scheduling, and 50 epochs of patience. It still overfit (train AUC 0.795 vs. test AUC 0.638) and underperformed the single-hidden-layer baseline. With ∼56,000 rows and 98 hand-crafted features, there is no representation to "learn"—the features are already engineered. Deep learning earns its keep when the inputs are raw (pixels, tokens, waveforms) and the model must discover structure. Here, gradient-boosted trees remain the industrial default for tabular data, and our results confirm why.

The SVM episode illustrates a lesson that textbooks often understate. At a 0.5 threshold, the calibrated linear SVM produced one positive prediction out of nearly 14,000. Its AUC was 0.6442—perfectly respectable. The model ranked patients correctly; it just did not know where to draw the line. After shifting the cutoff to 0.108, the SVM recovered 511 true positives and 2,721 false positives. Threshold selection is not a post-hoc nicety; it is as fundamental as model selection in imbalanced classification.

All four interpretable methods—decision tree feature importance, logistic regression absolute coefficients, XGBoost gain, and SHAP—converge on the same two dominant signals: discharge disposition (specifically, "discharged to home" as a protective factor) and number of prior inpatient visits (as a risk factor). Age, total prior visits, time in hospital, and circulatory diagnosis group appear

in most top-10 lists. This cross-method convergence is the strongest evidence we have that these features carry genuine predictive content rather than being artifacts of a particular splitting criterion or loss function.

The fairness audit revealed that the "Other" race group has the lowest recall (0.421) under both XGBoost and logistic regression, while African American patients fall slightly below Caucasian patients (0.513 vs. 0.547 for XGBoost). Because the gap pattern is consistent across model families with different inductive biases, we attribute it to data-level factors: unequal group sizes ("Other" has only 307 test rows), differences in baseline positive prevalence (ranging from 7.6% to 9.1% in training), and likely documentation and access disparities that we cannot disentangle from the available features. Deployment of any of these models would require ongoing monitoring under HIPAA-aligned governance, not a one-time audit.

We state our limitations bluntly. The data span 1999–2008; ICD-9 coding standards and treatment norms have changed. The binary target collapses "readmitted after 30 days" and "never readmitted" into a single class, losing clinical granularity. Removing deceased and hospice patients introduces survivorship bias. We use a single random train/test split with no external validation on a different hospital system. And we have no temporal features—no sequence of labs or vitals over the hospital stay—which likely accounts for a large fraction of the missing signal.

7 Conclusion and Future Work

We compared five classifiers on a leakage-aware, deduplicated version of the UCI Diabetes 130-US Hospitals dataset and found that XGBoost edged out the competition with 0.6505 AUC-ROC, while logistic regression delivered 0.6444 with a fraction of the complexity and the smallest train–test gap (0.013).

Our recommendation depends on context. For hospitals that need auditable coefficients—where a compliance officer or clinician must understand *why* a patient was flagged—logistic regression is the right starting point. Its coefficients are log-odds, its regularization path is well understood, and it barely overfits. For institutions that can invest in ensemble monitoring and SHAP-based explanations, XGBoost with a tuned threshold offers a small but consistent ranking advantage. Either way, the model should be paired with prospective validation, demographic monitoring, and an explicit cost-benefit analysis of the operating-point threshold. No model we tested—or have seen published—turns 30-day readmission prediction on administrative data into a solved problem.

Future work should explore SMOTE and other resampling strategies (we did not try them here and cannot claim they would help), Bayesian hyperparameter optimization to more efficiently search the XGBoost space, stacking ensembles that combine logistic and boosted heads, cost-sensitive learning with hospital-specific false-negative penalties, multiclass targets that separate ">30-day readmission" from "never readmitted," fairness-aware training objectives if disparities

persist after threshold adjustment, and external validation on a contemporary cohort with ICD-10 codes.

Declaration of AI Tool Usage

In accordance with Springer Nature's policy on the use of generative AI tools, we disclose the following.

Tools used. We used Claude (Anthropic) and ChatGPT-4 (OpenAI) at various stages of this project.

Code assistance. AI tools were used to help with boilerplate code structure (e.g., setting up cross-validation loops, configuring Keras callbacks), debugging syntax errors, and generating initial plotting functions for confusion matrices and ROC curves. All code was reviewed line by line, modified to match our experimental design, and tested by the authors. Hyperparameter grids, feature engineering decisions, and the preprocessing pipeline were designed by the authors based on domain reading and class material.

Writing assistance. AI was used for brainstorming section outlines, checking grammar, and suggesting paragraph transitions during early drafts. All text was substantially rewritten by the authors to reflect our own analysis, interpretation, and voice. Sentence-level phrasing, clinical interpretations, critical discussion of results, and all opinions expressed in this paper are our own.

What was not done by AI. All experimental design decisions—including the choice to enforce patient deduplication, the selection of class-weight strategies, the fairness audit design, and the threshold optimization methodology—were made entirely by the authors. Clinical interpretations, the identification of limitations, and the fairness discussion reflect our own reasoning.

Accountability. The authors take full responsibility for the accuracy and integrity of all content in this paper, including any errors.

Plagiarism statement. This work is original. All sources are cited. The Diabetes 130-US Hospitals dataset is publicly available from the UCI Machine Learning Repository under a CC BY 4.0 license. No text has been copied from other papers without proper attribution.

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